Network Programming

(SPG 0463)

Tasks Compilation:

Chapter 2 - Chapter 9

TRADE : CBS 1

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Chapter 2

Task 1 :

```
print("--- Task 1 ---")

print("Hello World!")

print("Hello Again")

print("I like typing this.")

print("This is fun.")

print("Yay! Printing.")

print("I'd much rather you 'not'.")

print('I "said" do not touch this.')

print("\n" + "="*20 + "\n")
```

Task 2 :

```
print("--- Task 2 ---")

# The input() function is used to get input from the user.

name = input("Enter your name: ")

# An f-string is used to easily embed the variable's value in the output
string.

print(f"Hello {name}")

print("\n" + "="*20 + "\n")
```

Task 3 :

```
print("--- Task 3 ---")

try:

    # We get input from the user and convert it to a floating-point
number.

    hours_str = input("Enter Hours: ")
```

```
    hours = float(hours_str)


    rate_str = input("Enter Rate: ")
    rate = float(rate_str)


    # Calculate the gross pay.
    pay = hours * rate


    # Print the final pay.
    print(f"Pay: {pay}")
except ValueError:
    # This block will run if the user enters text that is not a number.
    print("Error: Please enter numeric values for hours and rate.")
print("\n" + "="*20 + "\n")
```

Task 4 :

```
print("--- Task 4 ---")
# We ask for the user's age, height, and weight.
age = input("How old are you? ")
height = input("How tall are you? ")
weight = input("How much do you weigh? ")


# We use an f-string to print the summary sentence with the user's inputs.
print(f"So, you're {age} old, {height} tall and {weight} heavy")
print("\n" + "="*20 + "\n")
```

Chapter 3

Task 2 :

```
print("--- Task 1 ---")



# We ask for the hours and rate from the user.

hours_input = input("Enter Hours: ")

rate_input = input("Enter Rate: ")



try:

    # We try to convert the user's input into numbers (floating-point).

    # This is the line that might cause an error if the input isn't a
number.

    hours = float(hours_input)

    rate = float(rate_input)



    # If the conversion works, we calculate and print the pay.

    pay = hours * rate

    print("Pay:", pay)



except:

    # If the conversion in the 'try' block fails, this 'except' block
runs.

    # It prints an error message instead of crashing the program.

    print("Error, please enter numeric input")




print("\n" + "="*20 + "\n")
```

Task 3 :

```
print("--- Task 2 ---")


# Define the pay rates

normal_rate = 5

overtime_rate = 10


# Get hours worked from the user

hours_worked_input = input("Enter Hours: ")


try:

    # Convert the input to a number

    hours_worked = float(hours_worked_input)


    # Check if there was any overtime

    if hours_worked > 40:

        # Calculate regular pay for the first 40 hours

        regular_pay = 40 * normal_rate


        # Calculate the number of overtime hours

        overtime_hours = hours_worked - 40


        # Calculate the pay for the overtime hours

        overtime_pay = overtime_hours * overtime_rate


        # The total pay is the sum of regular and overtime pay
```

```
        total_pay = regular_pay + overtime_pay
    else:
        # If there is no overtime, the calculation is simple
        total_pay = hours_worked * normal_rate

    print("Pay:", total_pay)

except:
    # Handle non-numeric input for this task as well
    print("Error, please enter a numeric value for hours.")
```

Chapter 4

Task 1 :

```
print("--- Task 1 ---")


def computepay(hours, rate, overtime_rate):
    """Calculates pay, including overtime."""
    if hours > 40:
        regular_pay = 40 * rate
        overtime_hours = hours - 40
        overtime_pay = overtime_hours * overtime_rate
        total_pay = regular_pay + overtime_pay
    else:
        total_pay = hours * rate
    return total_pay


try:
    hours_input = input("Enter Hours: ")
    hours_worked = float(hours_input)


    # The rates are fixed as per the problem description
    normal_rate = 5
    overtime_rate = 10


    pay = computepay(hours_worked, normal_rate, overtime_rate)
    print("Pay:", pay)
except:
```

```
    print("Error, please enter a numeric value for hours.")


print("\n" + "="*20 + "\n")
```

Task 4 :

```
print("--- Task 4 ---")


def add(a, b):

    return a + b


def minus(a, b):

    return a - b


def multiply(a, b):

    return a * b


def divide(a, b):

    # It's good practice to check for division by zero.

    if b == 0:

        return "Error: Cannot divide by zero"

    return a / b


try:

    num1_input = input("Enter the first number: ")

    num2_input = input("Enter the second number: ")


    num1 = float(num1_input)
```

```
    num2 = float(num2_input)


    print(f"{num1} + {num2} = {add(num1, num2)}")
    print(f"{num1} - {num2} = {minus(num1, num2)}")
    print(f"{num1} * {num2} = {multiply(num1, num2)}")
    print(f"{num1} / {num2} = {divide(num1, num2)}")
except:
    print("Error: Please enter valid numbers.")


print("\n" + "="*20 + "\n")
```

Task 5 :

```
print("--- Task 5 ---")


def calculate_area(length, width):
    """Returns the area of a rectangle."""
    return length * width


try:
    length_input = input("Enter the length: ")
    width_input = input("Enter the width: ")


    length = float(length_input)
    width = float(width_input)


    area = calculate_area(length, width)
    print("The area of the rectangle is:", area)
```

```
except:
    print("Error: Please enter numeric values for length and width.")


print("\n" + "="*20 + "\n")
```

Task 8 :

```
print("--- Task 8 ---")


def show_employee(name, salary=9000):
    """Displays an employee's name and their salary."""
    print(f"Employee Name: {name}, Salary: {salary}")


# Calling the function with both name and salary
show_employee("Ben", 12000)


# Calling the function with only the name, so the default salary is used
show_employee("Jessica")
```

Chapter 5

Task 2 :

```
print("--- Task 2 ---")


# The list of numbers to check.
numbers = [56, 4, 50, 90, 76]
largest_number = None  # Initialize with None


print(f"List of numbers: {numbers}")


# We loop through each number in the list.
for number in numbers:
    # If largest_number is still None, it's the first time through the
loop.
    # So, we set the first number as the largest so far.
    if largest_number is None:
        largest_number = number
    # For all other numbers, we check if they are bigger than the current
largest.
    elif number > largest_number:
        largest_number = number


print("The largest number is:", largest_number)
print("\n" + "="*20 + "\n")
```

Task 3 :

```
print("--- Task 3 ---")
```

```
# The list for this task.
numbers_to_check = [2, 6, 8, 1, 12, 3]
smallest_number = None

print(f"List of numbers: {numbers_to_check}")

# Loop through each number.
for number in numbers_to_check:
    # If smallest_number hasn't been set yet, use the first number.
    if smallest_number is None:
        smallest_number = number
    # If the current number is smaller than our current smallest, update it.
    elif number < smallest_number:
        smallest_number = number

print("The smallest number is:", smallest_number)
print("\n" + "="*20 + "\n")
```

Task 4 :

```
print("--- Task 4 ---")

# Ask the user to enter some text.
user_input = input("Enter a string: ")

# A string containing all vowels (both lowercase and uppercase).
vowels = "aeiouAEIOU"
```

```
vowel_count = 0


# Loop through every character in the user's input string.
for character in user_input:
    # Check if the character is in our vowels string.
    if character in vowels:
        vowel_count = vowel_count + 1


print("The number of vowels is:", vowel_count)
print("\n" + "="*20 + "\n")
```

Task 5 :

```
print("--- Task 5 ---")


# We need the 'random' library to generate a random number.
import random


# Generate a random integer between 1 and 10.
secret_number = random.randint(1, 10)


print("I'm thinking of a number between 1 and 10.")


# This loop will continue until the user guesses correctly.
while True:
    guess_input = input("Take a guess: ")


    try:
```

```
        # Try to convert the user's guess into an integer.
        guess = int(guess_input)

        # Compare the guess to the secret number.
        if guess == secret_number:
            print("You got it! The number was", secret_number)
            break  # Exit the loop because the guess was correct.
        else:
            print("Wrong, try again!")

    except:
        # This 'except' block runs if the 'try' block fails
        # (e.g., if the user enters text instead of a number).
        print("Invalid guess. Please enter a number.")
```

Chapter 6

(Group 2 Lab Task) Set 2

Question :

Set 2 : str.find(sub[, start[, end]]) str.lstrip([chars]) str.replace(old, new[, count])

Hariz :

```
word = "pneumonoultramicroscopicsilicovolcanoconiosis"  # Real Lung Disease Btw 🤓
print("Word:", word)
print("Length of word:", len(word), "letters! 🤯")

#Cari A
print("\nFirst 'a' found at index:", word.find("a"))

#Cari Subtring, cth Volcano
print("Substring 'volcano' found at index:", word.find("volcano"))

#Cari benda huruf yg tkde dlm word
print("Searching for 'z'... Result:", word.find("z"))  # returns -1

#Cari A tapi start dari index, cth 13
print("Searching for 'a' starting from index 10:", word.find("a", 10))

#Cari in between
print("Searching for 'a' between index 12 until 30:", word.find("a", 12, 30))
```

Danial :

```
x = "pneumonoultramicroscopicsilicovolcanoconiosis"


print("\nOriginal word:", x)


#replace word

print("\nReplace 'micro' with 'mega': ")

print(x.replace("micro", "mega"))


#replace letter

print("\nReplace 7 of 'o' with 'A': ")

print(x.replace("o", "A", 7))
```

Putri :

```
# Perkataan asal

word = "pneumonoultramicroscopicsilicovolcanoconiosis"

print("🔤 Perkataan asal:", word)

print("📏 Panjang perkataan:", len(word), "huruf\n")


#1️⃣ Buang prefix 'pneumo' kat depan

print("👉 Buang 'pneumo' kat depan...")

remove_prefix = word.lstrip("pneumo")  # Buang semua huruf p,n,e,u,m,o kat depan sampai jumpa huruf lain

print("Hasil:", remove_prefix, "\n")


#2️⃣ Buang semua huruf vokal kat depan

print("👉 Buang huruf vokal (a, e, i, o, u) kat depan...")
```

```
remove_vowels = word.lstrip("aeiou")
print("Hasil:", remove_vowels)  # Dalam kes ni tak berubah sebab mula dengan 'p'

# Contoh kalau word start dengan vokal
print("\n🆚 Banding dengan word mula dengan vokal...")
word2 = "eupneumonoultramicroscopicsilicovolcanoconiosis"
print("Asal :", word2)
print("Hasil:", word2.lstrip("aeiou"))  # Huruf 'e' kat depan dibuang
```

Chapter 8

Task 1 :

```
print("--- Task 1 ---")


# Ask the user for their current shopping list as a single line of text.
initial_list_str = input("Enter your shopping list (e.g., Banana, Tomato, Carrot): ")


# Split the input string by commas to create a list of items.
# We'll also remove any extra spaces around each item.
shopping_list = []
for item in initial_list_str.split(','):
    shopping_list.append(item.strip())


# Ask for the new item to add.
new_item = input("Enter the new item: ")


# Add the new item to the end of the list.
shopping_list.append(new_item.strip())


# To display the list nicely like in the example, we join the items back into a string.
final_list_str = ", ".join(shopping_list)


print("Your new shopping list:")
print(final_list_str)
```

```
print("\n" + "="*20 + "\n")
```

Task 2 :

```
print("--- Task 2 ---")



# The starting list of words.

sentence_list = ['Saya', 'NWS2']



# Show the original list.

print(sentence_list)



# Ask the user for a word to insert.

word_to_insert = input("Enter a word: ")



# Calculate the middle index of the list.

# For a list with 2 items, the length is 2. 2 // 2 gives 1.

# The insert method places the new item at this index.

middle_index = len(sentence_list) // 2



# Insert the new word at the calculated middle index.

sentence_list.insert(middle_index, word_to_insert)



# Print the final, modified list.

print(sentence_list)
```

Chapter 9

Task 1 :

```
print("--- Task 1 ---")



# Ask the user to enter a word.

word = input("Enter a word to count its characters: ")



# Create an empty dictionary to store the counts.

character_counts = {}



# Loop through each character in the input word.

for character in word:

    # .get(key, default) is a useful dictionary method.

    # It tries to get the value for 'character'. If the character isn't a
key yet,

    # it returns the default value (0 in this case).

    # We then add 1 to the count and store it back in the dictionary.

    character_counts[character] = character_counts.get(character, 0) + 1



print("Character counts:")

print(character_counts)



print("\n" + "="*20 + "\n")
```

Task 2 :

```
print("--- Task 2 ---")
```

```
# The dictionary maps a choice (the key) to a character name (the value).
character_choices = {
    'A': 'Ultraman',
    'B': 'Power Rangers',
    'C': 'Masked Rider',
    'D': 'Power Puff Girl'
}

# Display the options to the user.
print("What characters that would you like to be")
print("Type A for Ultraman")
print("Type B for Power Rangers")
print("Type C for Masked Rider")
print("Type D for Power Puff Girl")

# Get the user's choice. We use .upper() to make it case-insensitive (so 'b' works like 'B').
choice = input("Choice: ").upper()

# Check if the user's choice is a valid key in our dictionary.
if choice in character_choices:
    # If it is, print the corresponding value.
    print("You've selected", character_choices[choice])
else:
```

```
    # If the choice is not a valid key, print an error message.
    print("Invalid choice. Please enter A, B, C, or D.")
```